

A Dyadic-Digit Extension Showing that $AD(0,1)$ Is Not Maximal

Candidate full negative answer to Problem 3.9 of arXiv:2002.00590

Automated mathematics run `fa_banach.001`, lane 15

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Status. Candidate full negative answer, likely valid, requiring expert review.

1 Source problem

Ber, Huang, Kudaybergenov, and Sukochev prove that the classical derivative has no dyadic-translation-invariant extension to all of $S(0,1)$ and ask whether the algebra of approximately differentiable functions is nevertheless maximal among subalgebras admitting such an extension [1, Problem 3.9]. Their question is:

Is the algebra $AD(0,1)$ a maximal subalgebra in $S(0,1)$ admitting a translation-invariant derivation as an extension of $\frac{d}{dt} : D(0,1) \rightarrow S(0,1)$?

The complete source statement appears on source PDF p. 13:

derivation $\delta : \mathcal{A} \rightarrow S(0,1)$ commuting with G . □

The following result is an immediate consequence of Theorem 3.2, which shows that the translation invariance property of $\frac{d}{dt}$ is not shared by its extension.

Corollary 3.8. *The differential operator $\frac{d}{dt}$ has no translation-invariant extension to the algebra $S(0,1)$.*

Theorem 3.2 holds for a subalgebra $(\mathcal{A}, \|\cdot\|_{\mathcal{A}})$ of $S(0,1)$ in which simple functions are dense. However, simple functions are not dense in $(AD(0,1), \rho)$. By Theorem 3.2, for any complete symmetric Δ -normed subalgebra $(\mathcal{A}, \|\cdot\|_{\mathcal{A}})$ of $S(0,1)$ in which simple functions are dense, there exists no derivation from $\mathcal{A}$ into $S(0,1)$ commuting with all dyadic-rational translations on $(0,1)$. Also, recall that it is shown in [8] that the algebra $AD(0,1)$ is the maximal subalgebra of $S(0,1)$ admitting unique extension of $\frac{d}{dt} : D(0,1) \rightarrow S(0,1)$. It is interesting to drop the “symmetric Δ -normed” assumption and consider the following problem.

Problem 3.9. *Is the algebra $AD(0,1)$ a maximal subalgebra in $S(0,1)$ admitting a translation-invariant derivation as an extension of $\frac{d}{dt} : D(0,1) \rightarrow S(0,1)$?*

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2 Definitions

Identify $(0,1)$ modulo endpoints with the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, equipped with Lebesgue measure μ . Write $S(0,1) = L^0(\mathbb{T})$ modulo almost everywhere equality. The algebra $AD(0,1)$ consists of classes with

a finite approximate derivative almost everywhere; its derivation is denoted ∂_{AD} . It is a regular, integrally closed unital $*$ -subalgebra of $S(0,1)$ and contains every idempotent [2, Section 3].

For a dyadic rational r , let

$$(\alpha_r f)(t) = f(\{t - r\}).$$

The derivation ∂_{AD} commutes with every α_r [1]. An element $a \in S(0,1)$ is *weakly transcendental* over a subalgebra A if $a \neq 0$ and no nonzero localization ea , where $e \leq \text{supp}(a)$ is an idempotent, is integral over A .

3 Proof intuition

Dyadic rotations preserve the infinite binary tail of a point and only permute a finite binary prefix. This suggests adjoining a function that depends strongly on every binary digit: all of its dyadic-translation increments will then be finite-valued, hence invisible to the approximate derivative. We use

$$h(t) = \sum_{n \geq 1} (3/4)^n b_n(t).$$

The weights decay too slowly relative to the spatial scale 2^{-n} . On the half-set where the n th digit is zero, moving by 2^{-n} creates slope $(3/2)^n$. This occurs asymptotically on half of every positive-measure set, so no nonzero localization of h is approximately differentiable. Integral closure makes h weakly transcendental, and the standard extension theorem then permits the prescription $\delta(h) = 0$. The finite-valued translation increments ensure that this extension remains translation invariant.

4 The counterexample

Choose, outside the null set of ambiguous binary expansions,

$$t = .b_1(t)b_2(t)b_3(t)\dots \quad (b_n(t) \in \{0,1\}), \quad h(t) = \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n b_n(t).$$

The series converges uniformly and defines a bounded real measurable function. Its support equals 1 modulo null sets.

Lemma 1 (Translation quotients of AD functions). *If $f \in \text{AD}(0,1)$, then, as $s \rightarrow 0$,*

$$Q_s f(t) := \frac{f(t+s) - f(t)}{s} \longrightarrow \partial_{\text{AD}} f(t)$$

in measure on $\mathbb{T}$.

Proof. Every $f \in \text{AD}(0,1)$ has a representation

$$f = \sum_{j \geq 1} \mathbf{1}_{E_j} g_j,$$

where (E_j) is a measurable partition modulo null sets and each $g_j \in C^1[0,1]$ [2, Proposition 3.4]. Given $\varepsilon > 0$, choose J so that $\mu(\bigcup_{j \leq J} E_j) > 1 - \varepsilon$. Translation continuity in measure gives

$$\mu(E_j \triangle (E_j - s)) \longrightarrow 0 \quad (j \leq J).$$

Except on a set of measure at most $\varepsilon + o(1)$ (and the wrap-around set of measure $|s|$), a point and its translate lie in the same one of the first J pieces. There $Q_s f = Q_s g_j$, which converges uniformly to g'_j for each of the finitely many j . On density points of E_j , $\partial_{\text{AD}} f = g'_j$. Since ε is arbitrary, the asserted convergence in measure follows. $\square$

Lemma 2 (No localized AD part). *For every measurable $E \subset \mathbb{T}$ with $\mu(E) > 0$, $\mathbf{1}_E h \notin \text{AD}(0, 1)$. Consequently h is weakly transcendental over $\text{AD}(0, 1)$.*

Proof. Put $A_n = \{t : b_n(t) = 0\}$ and $s_n = 2^{-n}$. Apart from binary endpoints, if $t \in A_n$, adding s_n changes the n th digit from zero to one, creates no carry, and leaves all other digits unchanged. Hence

$$\frac{h(t + s_n) - h(t)}{s_n} = \frac{(3/4)^n}{2^{-n}} = \left(\frac{3}{2}\right)^n \quad (t \in A_n). \quad (1)$$

Let $r_n = 1 - 2b_n$. The Rademacher functions (r_n) are orthonormal in $L^2(\mathbb{T})$, hence converge weakly to zero. Therefore

$$\mu(E \cap A_n) = \frac{1}{2}\mu(E) + \frac{1}{2} \int_E r_n d\mu \longrightarrow \frac{1}{2}\mu(E).$$

Translation continuity of measurable sets also gives

$$\mu(E \cap (E - s_n) \cap A_n) \longrightarrow \frac{1}{2}\mu(E).$$

On this last set both t and $t + s_n$ lie in E , so by (1) the quotient $Q_{s_n}(\mathbf{1}_E h)$ equals $(3/2)^n$. Thus these quotients are not tight in measure: for every fixed M , their absolute value exceeds M on a set whose measure tends to $\mu(E)/2$. Lemma 1 rules out $\mathbf{1}_E h \in \text{AD}(0, 1)$.

Since $\text{AD}(0, 1)$ is integrally closed in $S(0, 1)$ [2], any element integral over $\text{AD}(0, 1)$ already belongs to $\text{AD}(0, 1)$. The preceding paragraph therefore says that no nonzero idempotent localization of h is integral over $\text{AD}(0, 1)$. Since $\text{supp}(h) = 1$, this is precisely weak transcendence. $\square$

Lemma 3 (Finite-valued dyadic cocycle). *For every dyadic rational $r = m/2^N$, the function $c_r := \alpha_r(h) - h$ is finite-valued. In particular, $c_r \in \text{AD}(0, 1)$ and $\partial_{\text{AD}} c_r = 0$.*

Proof. Write $t = (j + u)/2^N$, where $j \in \{0, \dots, 2^N - 1\}$ and $0 \leq u < 1$. The first N binary digits encode j , while the later digits are the binary digits of u . Translation by $m/2^N$ replaces j by $j - m$ modulo 2^N and leaves u unchanged. Thus the infinite tails in $\alpha_r(h)$ and h cancel. Their difference depends only on j and has at most 2^N values. Every finite-valued measurable function belongs to $\text{AD}(0, 1)$ with approximate derivative zero. $\square$

Theorem 1 (Negative answer to Problem 3.9). *The algebra $\text{AD}(0, 1)$ is not maximal among subalgebras of $S(0, 1)$ admitting a dyadic-translation-invariant derivation extending $d/dt : D(0, 1) \rightarrow S(0, 1)$.*

Proof. Ber–Chilin–Sukochev’s weak-transcendental extension theorem states the following. If A is a regular subalgebra of a complete commutative regular algebra S , A contains every idempotent of S , $\partial : A \rightarrow S$ is a derivation, and $a \in S$ is weakly transcendental over A , then for every $z \in S$ with $\text{supp}(z) \leq \text{supp}(a)$ there is a derivation on $A[a]$ extending ∂ and taking a to z [3, Proposition 3.7].

Apply this with

$$S = S(0, 1), \quad A = \text{AD}(0, 1), \quad \partial = \partial_{\text{AD}}, \quad a = h, \quad z = 0.$$

The structural hypotheses on A are recalled above, and Lemma 2 supplies weak transcendence. We obtain a derivation

$$\delta : \mathcal{B} := \text{AD}(0, 1)[h] \longrightarrow S(0, 1), \quad \delta|_{\text{AD}} = \partial_{\text{AD}}, \quad \delta(h) = 0.$$

Lemma 2 also gives $h \notin \text{AD}(0, 1)$, so $\mathcal{B} \supsetneq \text{AD}(0, 1)$. Because h is real, $\mathcal{B}$ is a $*$ -subalgebra.

Lemma 3 gives $\alpha_r(h) = h + c_r$ with $c_r \in \text{AD}(0, 1)$, so $\alpha_r(\mathcal{B}) = \mathcal{B}$. The two derivations $\delta\alpha_r$ and $\alpha_r\delta$ agree on $\text{AD}(0, 1)$ because ∂_{AD} is translation invariant. They also agree on h , since

$$\delta(\alpha_r h) = \delta(h + c_r) = 0 + \partial_{\text{AD}} c_r = 0 = \alpha_r(\delta h).$$

They consequently agree on the algebra generated by $\text{AD}(0, 1)$ and h . Thus δ commutes with every dyadic translation and extends the classical derivative, completing the negative answer. $\square$

5 Verification and limitations

The proof uses three independently checkable ingredients: the C^1 -piece representation of AD , the exact Rademacher half-density argument, and Proposition 3.7 of [3]. Exact finite binary arithmetic checks in `code/verifier.py` test both the slope identity and the finite-prefix cocycle; they are sanity checks, not proof.

This construction extends the derivation only to the proper intermediate algebra $\text{AD}(0, 1)[h]$, not to all of $S(0, 1)$. It therefore does not contradict Corollary 3.8 of [1]. Nor is the enlarged algebra claimed to satisfy the complete symmetric Δ -norm hypotheses of their nonexistence theorem.

6 Novelty check

Bounded searches on 11 August 2026 covered the four lightweight run indexes; the exact Problem 3.9 sentence; the source title and arXiv id; combinations of “ $\text{AD}(0, 1)$ ”, “maximal subalgebra”, “translation-invariant derivation”, “dyadic translation”, “weakly transcendental”, and “binary digit”; the 2023 journal version; later citation variants; and the 2023 paper on the isomorphism between $S(0, 1)$ and $\text{AD}(0, 1)$. No explicit resolution or matching construction was found. Novelty confidence is moderate and provisional.

7 Human-review recommendation

Verify Lemma 1, especially the passage from the countable C^1 partition to convergence in measure; verify that Lemma 2 matches the precise definition of weak transcendence; and check the hypotheses of [3, Proposition 3.7]. The remaining translation calculation is finite binary arithmetic.

References

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